
Bayer & myHealthteam

— MHT Menopause Community —
Review - Climara

AGENDA

- Approach
- Solution Overview
- Insights and Findings
- Additional Considerations & Next Steps
- Q & A
- Appendix

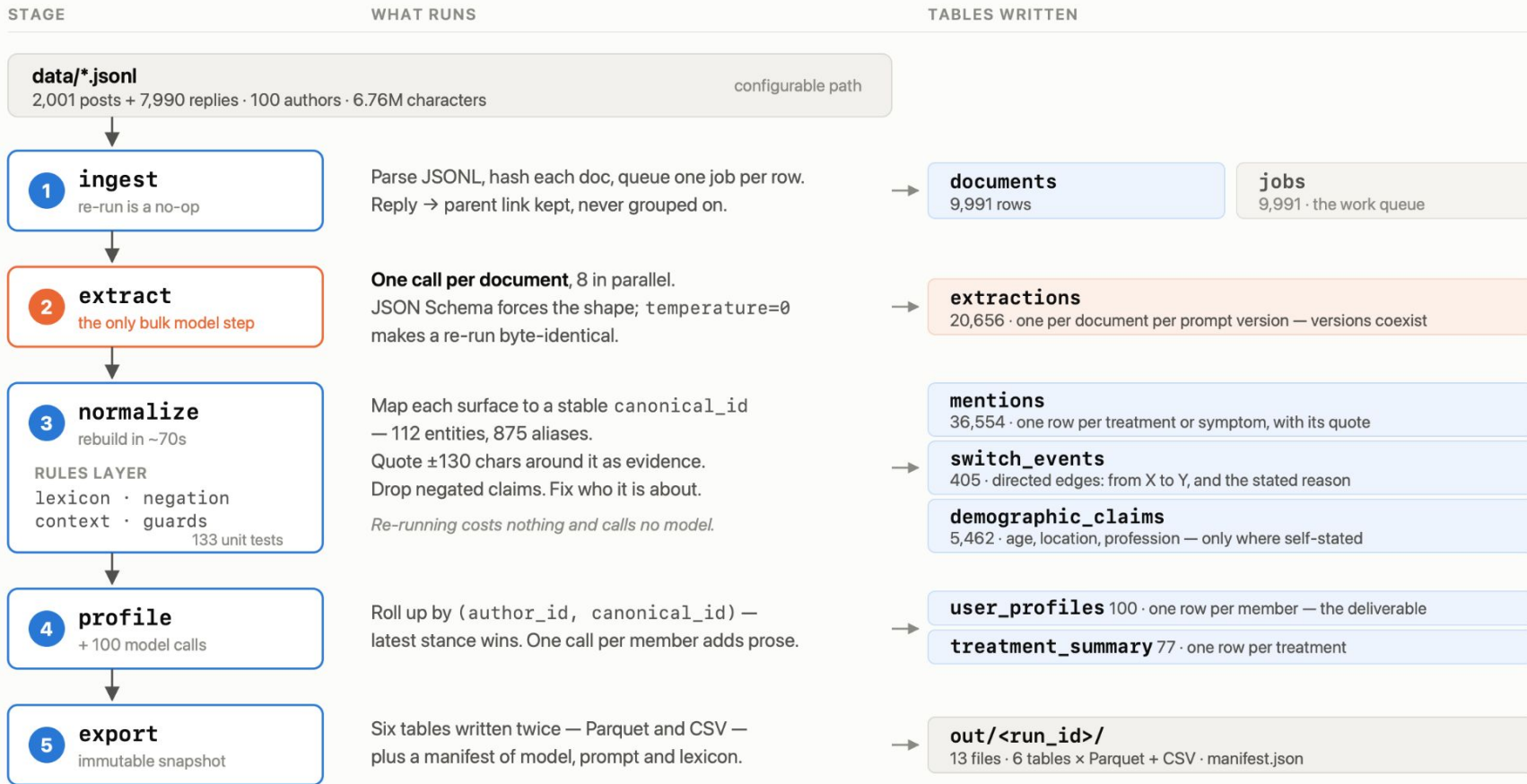
Approach

- **Goal:** Extract a set of analytical data products to support Climara patient community analyses
- Data
 - 2 [*.jsonl] files with ~10,000 MyHealthTeam Menopause Community Posts and Replies
 - ~2,000 posts
 - ~8,000 replies (*to provided parent posts)
 - Included data attributes: Post/Reply - body, author, unique identifier and creation date
- Solution
 - Overview

Forum text → a per-member structured dataset

model deterministic

One model call per document. Everything after it is deterministic and re-runnable for free.



Approach

- Assumptions/Caveats
 - User-generated content (community forum posts) is assumed to be factual
 - A “Climara patient” is defined as an author of a post or reply that makes a *self-reference to having taken Climara
 - * Identification of self-reference is done by LLM named-entity recognition prompt
 - “Sentiment” is categorized via LLM sentiment classifier prompt
- Solution
 - Performance Overview

Guided Review Questions

(?) Who is the Climara patient in the community?

(?) How are community members feeling about Climara?

(?) What other treatments are members using?

What is measured

9,991 documents · llama3.1:8b · prompt v7 · lexicon 46a8b905c4eb

Documents extracted 9,991 / 9,991

Schema-valid JSON returned 100.00%

Named-entity recall — 34 entities, 6,813 pairs 95.6%

the 34 of 113 lexicon entities with a single distinctive name that occurs in this corpus

Names in neither the document nor its parent 30

0.47% of the 6,320 mentions of scored entities; all 30 are one entity, black_cohosh

Negation & demographic controls 8 / 8

Mentions mapped to a lexicon entity 88.5%

Mentions carrying a quoted context 98.5%

Surface forms collapsed to entities 3,159 → 110 (28.7 : 1)

Same 50 documents, extracted 3x — agreement 1.000 Jaccard

raw surface forms score 0.997 on the same sample — decoding is greedy at temperature=0

Produced by make_evaluate and make_stability.

Insights and Findings

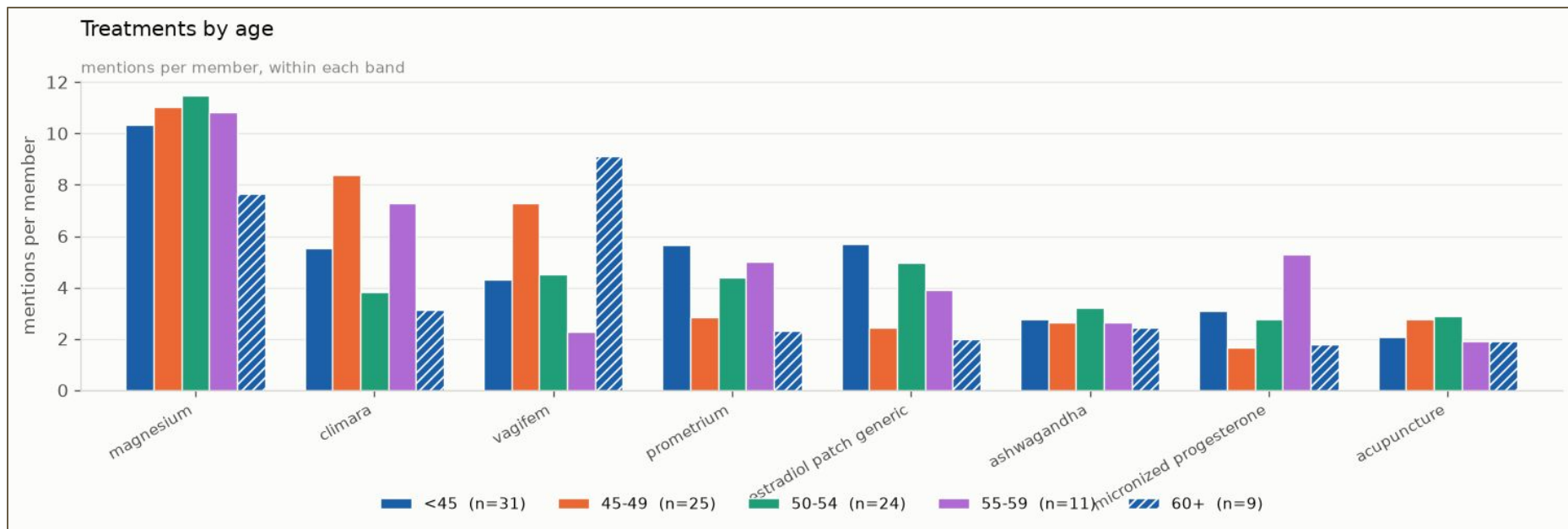
- Who is the MHT Menopause Patient Community?

Members	100
Documents	9,991 (2,001 posts · 7,990 replies)
Active	Oct 2025 – May 2026
Documents per member	98 (74–121)
Age	49 (38–64)
Treatments named per member	20 (11–33)
Symptoms named per member	23 (15–29)
Switch events per member	3 (0–11)
The last five rows show median (range) across the 100 members.	

Cut	Group	Members
Age	<45	31
	45–49	25
	50–54	24
	55–59	11
	60+	9
Location	United States	52
	Canada	21
	Australia	14
	UK & Ireland	12
Profession	Corporate & professional	41
	Healthcare	19
	Creative	16
	Service & hospitality	13
	Education & other	9
Climara	Took Climara	51
	Did not take it	49

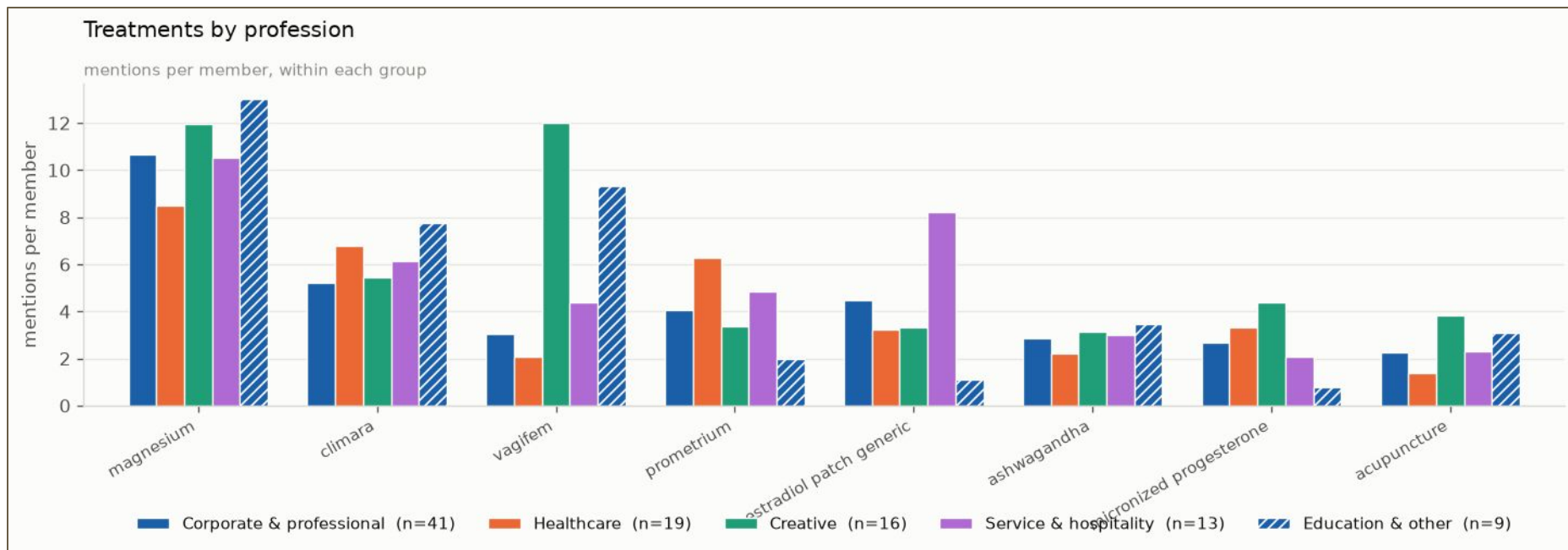
Insights and Findings

- Who is the MHT Menopause Patient Community?



Insights and Findings

- Who is the MHT Menopause Patient Community?

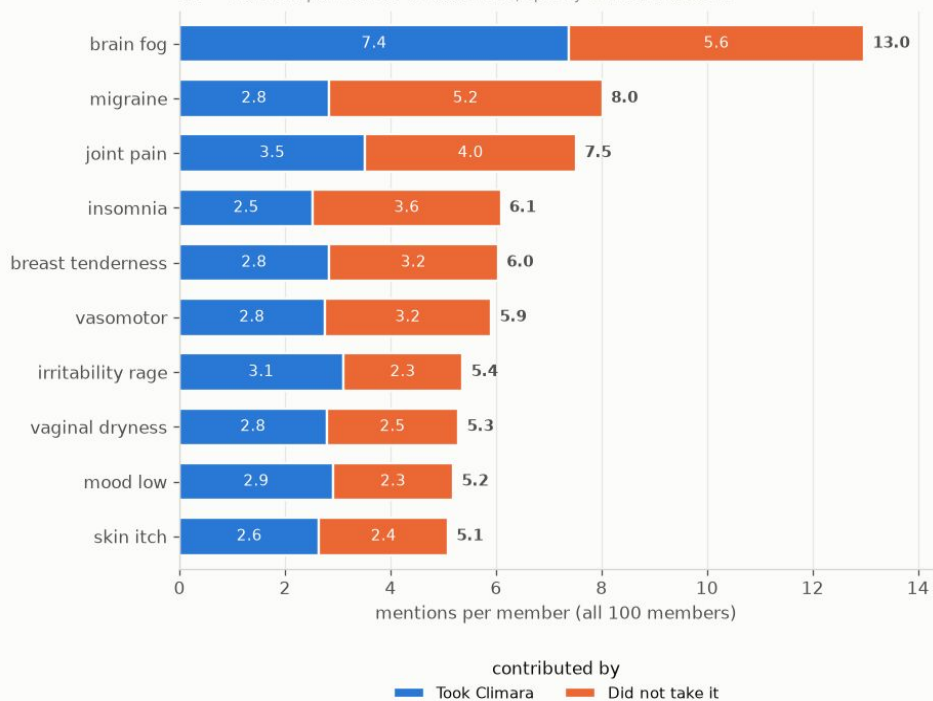


Insights and Findings

- What is the Climara Patient Community experience?

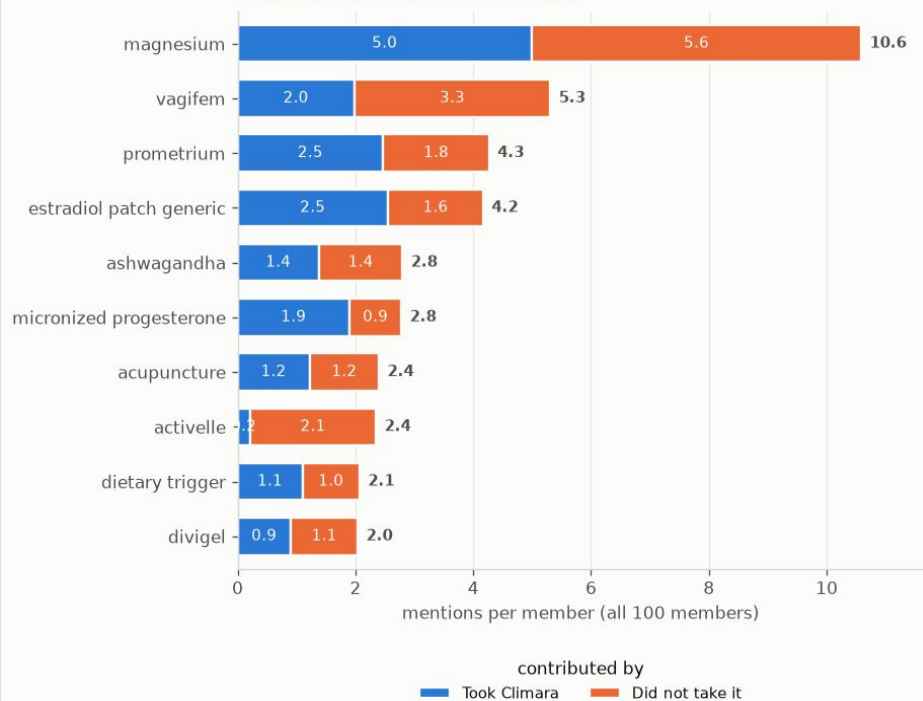
What the community lives with

bar = mentions per member across all 100, split by who contributed it



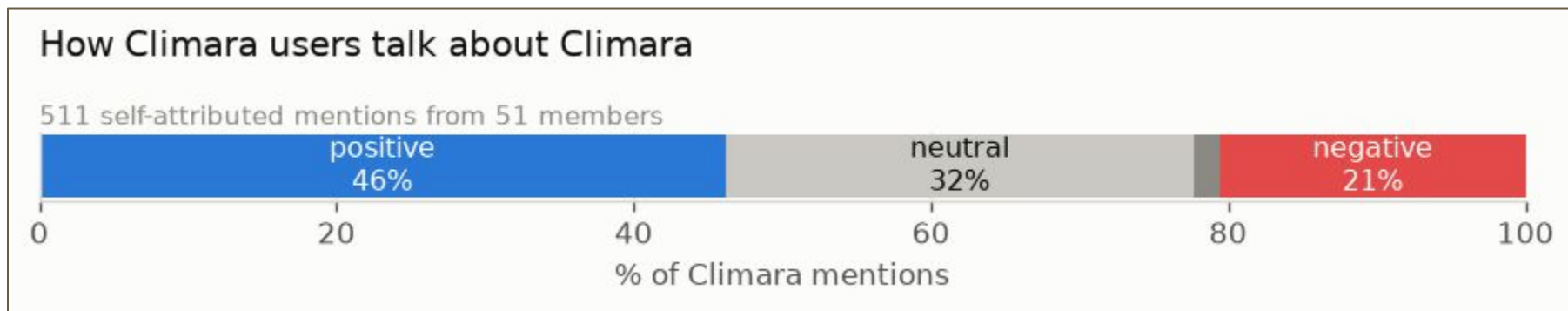
What they take for it

umbrella entities and Climara itself excluded



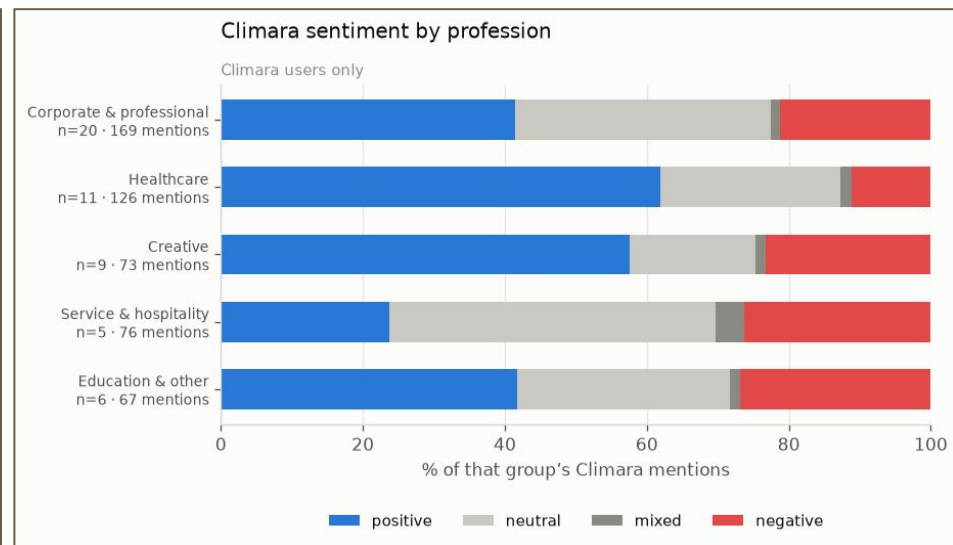
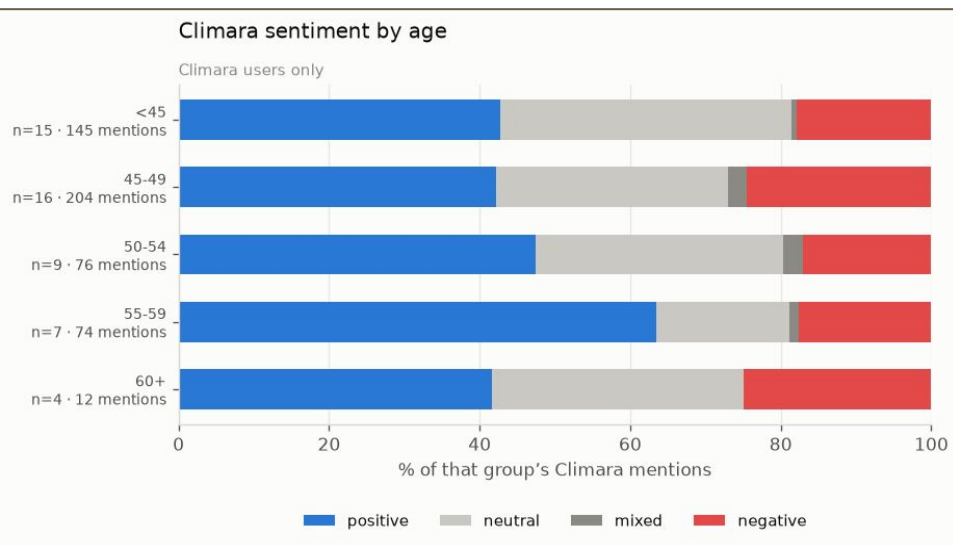
Insights and Findings

- How does the MHT Climara Patient Community feel about the treatment?

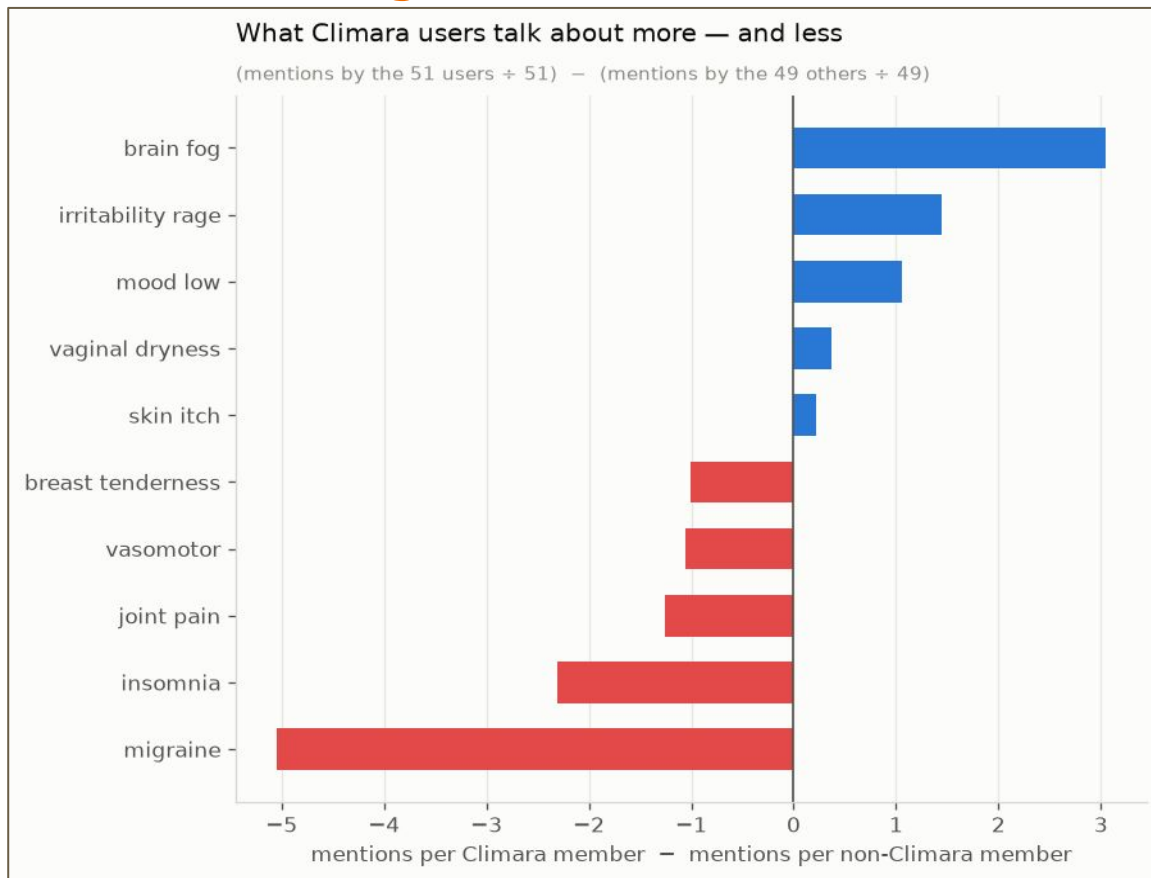


Insights and Findings

- How does the MHT Climara Patient Community feel about the treatment?



Insights and Findings



Insights and Findings

- What treatments do the MHT Patient Community switch *TO* and *FROM*?

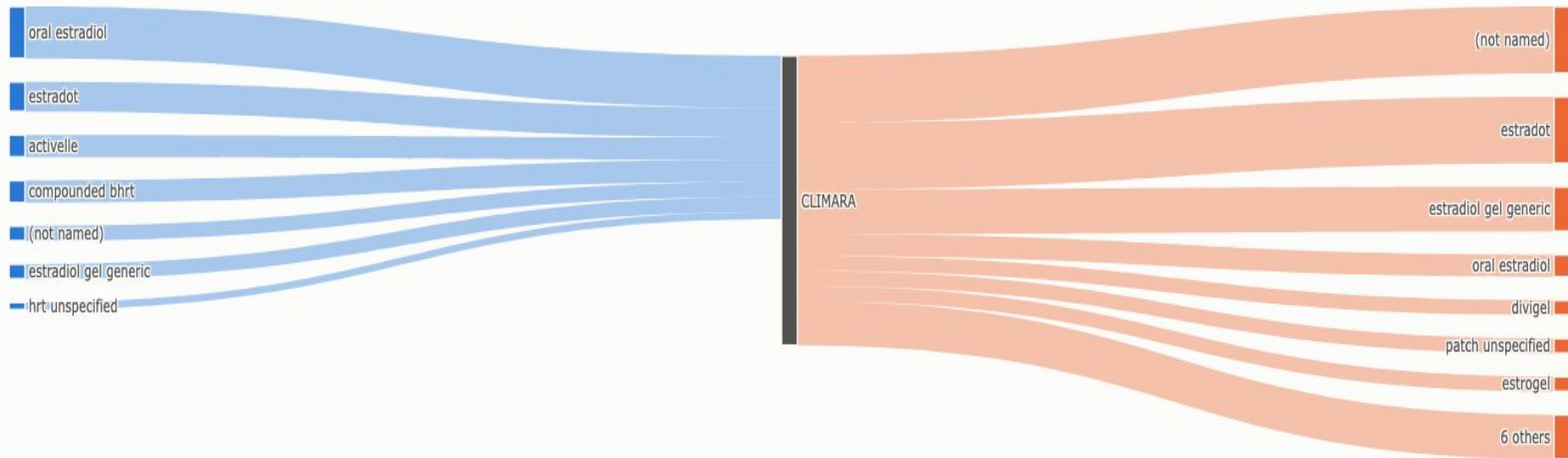
Switches between treatment classes (159 of 338 events where both sides resolve to a class)								
from ↓ to →	Patch	Gel / spray	Oral estrogen	Oral combined	Progestogen	Local / vaginal	Local DHEA	Compounded
Patch	17	55	4	1	3	2		1
Gel / spray	9	1				2		
Oral estrogen	12					1		
Oral combined	10				1	2		
Progestogen	1				13	2		
Local / vaginal	3	1	1			7	2	
Compounded	6				2			

Insights and Findings

- What treatments do the MHT Climara Patient Community switch *TO* and *FROM*?

Switching into and out of Climara

22 events in from 18 members - 39 out from 25 members



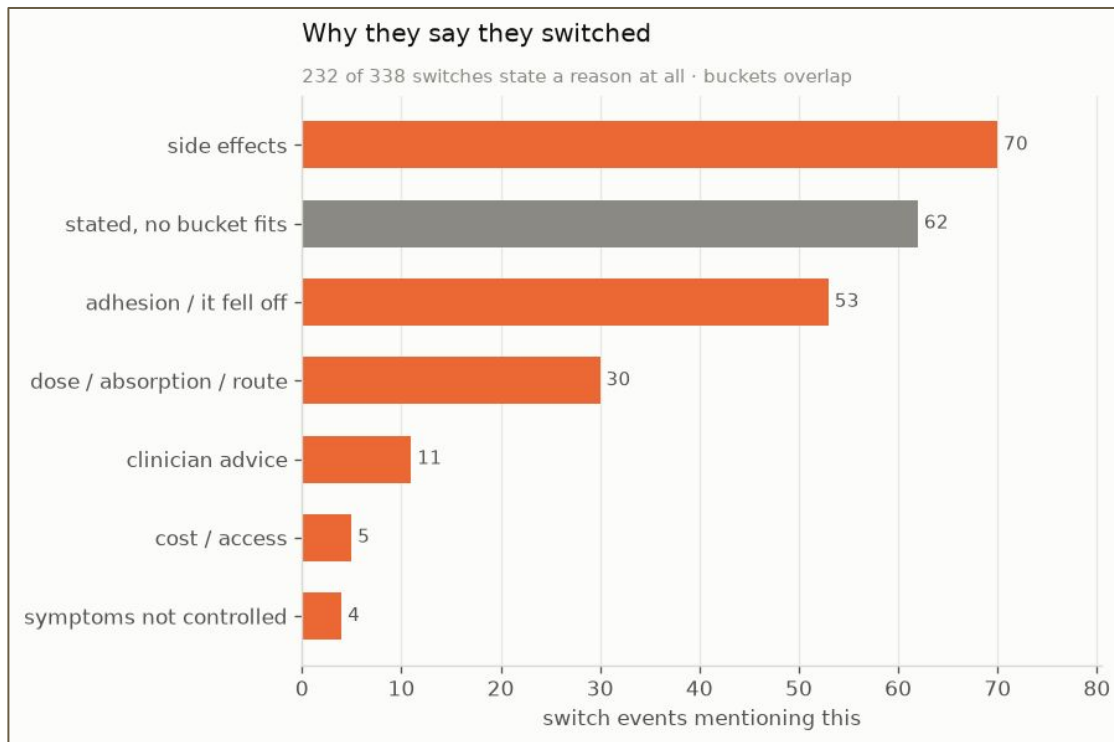
Insights and Findings

- What treatments do the MHT Climara Patient Community switch *TO* and *FROM*?

Climara switch flows			
Treatment	to Climara	from Climara	net
oral estradiol	7	3	+4
activelle	3	0	+3
compounded bhrt	3	0	+3
divigel	0	2	-2
estrogel	0	2	-2
patch unspecified	0	2	-2
estradiol gel generic	2	6	-4
estradot	4	9	-5
(not named)	2	9	-7
7 others, one move each	1	6	-5
TOTAL	22	39	-17

Insights and Findings

- Why does the MHT Menopause Patient Community switch treatments?



Additional Considerations & Next Steps

- Scaling for Production
 - Keep as LLM/agent-based → vLLM on cloud compute ⇒ 37 million posts for < ~\$5,000, ~3-8 days
- Validation
 - More human review if given more time
 - “lexicon” configuration vs. MHT Medical Ontology Reference database
 - “domain” configuration vs. MHT internal knowledge reference(s)
- Analytical Data Products
 - Switch Events - Refine prompt; more quality measures/tests
 - Enrichment - Try to extract additional output:
 - Efficacy
 - Dosage
 - Ontology mappings

Q & A

Thank you for your time! What questions do we have?

APPENDIX

Sentiment - Member Comments

=== POSITIVE ===

- ...I'm not going to offer anything except – yes. This is grief and it's real and it deserves to be called what it is. I'm on the Climara patch now and it's helped the rage a bit but it hasn't fixed the feeling of being a stranger to myself, so I know that part isn't just...
- I made exactly this transition about two years ago – Vagifem plus added Climara patch, and then Prometrium. First month: breast tenderness yes, quite noticeable, lasted roughly six weeks for me. Bloating was a...
- ...me. Also did the iron supplementing cause any stomach issues? I'm already anxious about adding anything new on top of the Climara. Really glad you found something that works though, 1-2 nights vs 5-6 is massive.

=== NEGATIVE ===

- temple hair loss getting worse even on the patch – anyone else?? okay so i've been on the Climara patch for almost 8 months now and i genuinely thought it would help slow down the hair loss at my temples but honestly?? it seems like...
- ...exhausting, especially doing detailed design work through daily headaches💛 I stopped briefly about two years before starting my Climara patch and honestly the muscle aches were the thing that surprised me most – I didn't even realize HRT had been helping with them until...
- ...good evidence for meibomian gland dysfunction. I'm filing this away because my allergies have been flaring since starting the Climara patch and my eyes are part of it.

Switch Events - Member Comments

22 of 338 switch events met the selection criteria; showing 5

SELECTION CRITERIA: Must carry a first-person movement construction, both endpoints must resolve to named products, and the surrounding text must not attribute the move to someone else

vagifem → **imvexxy** [applicator faff]

"Switched from Vagifem to Imvexxy a while back mostly because of applicator faff tbh"

activelle → **climara** [reduce first-pass hepatic effects]

"switching from oral to transdermal estrogen"

estradot → **estradiol gel generic** [thinking transdermal might be gentler]

"Switched to estrogen gel thinking transdermal might be gentler."

prometrium → **mirena**

"switching to a Mirena IUD for the progestogen component"

vagifem → **intrarosa** [anecdotally the urgency did settle a bit more]

"Switched to Intrarosa when my compounding clinic suggested it"

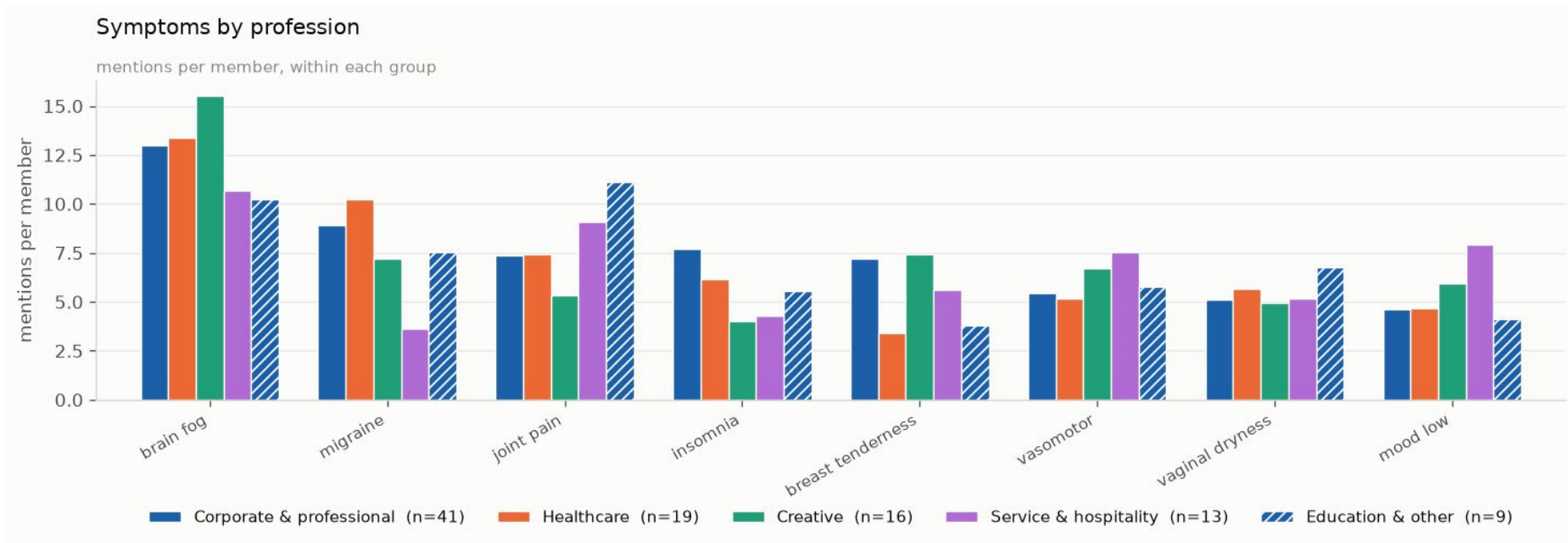
User “switch event” - Reason Text Mapping

```
# The stated reasons, in members' own words. `reason` is free text with no
# canonical form - `adhesion issues` (47) and `adhesive issues` (6) are separate
# strings - so it is bucketed by regex rather than grouped. The patterns were
# widened against the actual unmatched values: efficacy is written as "breaking
# through", "not responding" and "wasn't resolving", never as "not working".
```

```
BUCKETS = {  
    'side effects': (r'tender|bleed|migraine|headache|nausea|bloat|irritat|rash|mood'  
        r'|adverse|side effect|ache|hair'),  
    'adhesion / it fell off': r'adhes|stick|peel|fall|fell off|came off|applicator',  
    'dose / absorption / route': (r'dose|absor|level|consistenc|strength|titrat|route'  
        r'|transdermal|systemic|peak|trough|hepatic'),  
    'clinician advice': r'doctor|gp|specialist|consultant|provider|nams|advice',  
    'cost / access': r'cost|price|insur|cover|afford|supply|shortage',  
    'symptoms not controlled': (r'not work|didn.t work|wasn.t work|no relief|ineffect'  
        r'|stopped work|nothing|breaking through|not respond'  
        r'|wasn.t respond|weren.t resolv|wasn.t resolv|got worse'  
        r'|unsustainable|wasn.t right'),  
}
```

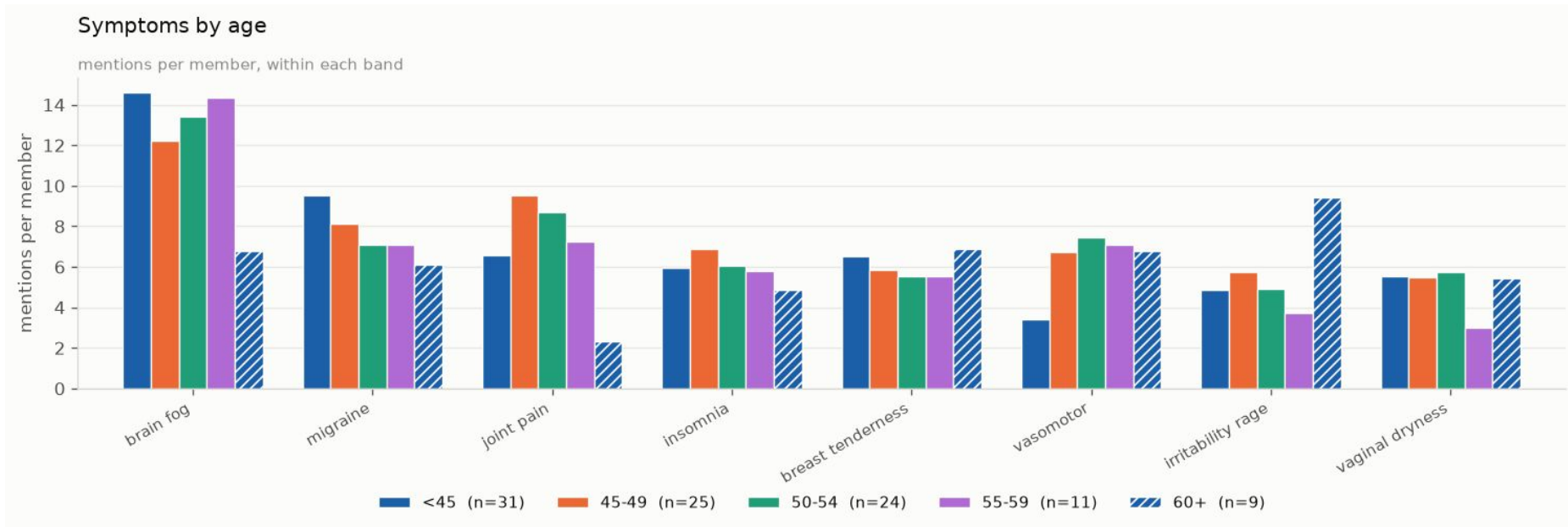
Insights and Findings

- Who is the MHT Menopause Patient Community?



Insights and Findings

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How accurate is it?

Every figure below comes from one command — `make evaluate` — and needs no human annotation, so it re-runs in seconds on any version.

95.6%

named-entity recall

6,515 of 6,813 document-entity pairs

100%

parse success

9,991 of 9,991 documents · zero failures

8 / 8

negation controls pass

4 that must reject, 4 that must accept

0.47%

names found nowhere in the text

30 of 6,320 mentions of scored entities

WHAT IS MEASURED

Documents extracted	9,991 / 9,991
Schema-valid JSON returned	100.00%
Named-entity recall — 34 entities	95.6%
Names in neither the document nor its parent	30
Negation & demographic controls	8 / 8
Mentions mapped to a lexicon entity	88.5%
Mentions carrying a quoted context	98.5%
Surface forms collapsed to entities	3,159 → 110 (28.7 : 1)

Same 50 documents, extracted 3x — agreement on entities

1.000 Jaccard

perfect. Greedy decoding at temperature=0 — determinism is a setting, not a hope.

Raw surface forms already score 0.997, so this measures determinism, not the value of canonicalisation — that case rests on the 28.7 : 1 collapse above.

Where it is weakest

Telehealth brands read as ordinary words in running text: `evernow` 71%, `mid_i_health` 71%.

All 30 unsupported names are one entity — `black_cohosh`, the known decoy case.

HOW RECALL IS SCORED WITHOUT ANNOTATION

A **named** entity has no casual synonym. You either write “Climara” or you do not refer to it at all — unlike “the patch”, “my hormones” or “exercise”, which have a dozen phrasings.

So the gold standard is a string search:

for every document whose own text contains the name, did the extraction emit that entity? 6,813 pairs, zero labelling.

The entity list is derived from the lexicon, not hand-written, so swapping `lexicon.yml` for a different drug or disease carries the evaluation with it. Nothing here is specific to this corpus.

WHAT IS NOT MEASURED

Three model-derived fields carry no accuracy number.

- `sentiment` — is “positive” about this treatment correct?
- `switch direction` — is from/to the right way round?
- `stance` — current vs past vs considering

Each needs human annotation. Recall and the controls do not, which is exactly why those are the numbers on this slide.

REPRODUCE IT

`make evaluate`

Prints every figure above as a table. No model calls — it reads the stored extractions, so it is free and takes seconds.

llama3.1:8b · prompt v7 · lexicon 46a8b905c4eb
9,991 documents · 5h45m · 0.48 docs/s on an M5 Max

The stability figure comes from `make stability`, which re-extracts a sample several times — the only check here that spends model calls (150 for the run above).

named-entity recall (deterministic gold: the name appears in the document's own text)

entity	docs	found	recall	emitted w/o text
magnesium	2507	2458	98%	0
vagifem	751	746	99%	0
climara	657	646	98%	0
prometrium	613	595	97%	0
ashwagandha	507	462	91%	0
activelle	290	286	99%	0
estradot	264	255	97%	0
divigel	246	243	99%	0
evernow	239	169	71%	0
omega3	172	149	87%	0
mirena	96	92	96%	0
black_cohosh	83	64	77%	30
quercetin	41	39	95%	0
utrogestan	38	36	95%	0
minoxidil	37	33	89%	0
estrogel	28	28	100%	0
imvexxy	25	24	96%	0
turmeric	25	24	96%	0
midi_health	24	17	71%	0
collagen	23	17	74%	0
estring	22	21	95%	0
biotin	21	20	95%	0
l_theanine	20	18	90%	0
intrarosa	16	14	88%	0
ospemifene	16	12	75%	0
d_mannose	15	14	93%	0
ovestin	8	8	100%	0
creatine	8	7	88%	0
spironolactone	7	6	86%	0
probiotics	5	3	60%	0
gabapentin	4	4	100%	0
estradiol_spray	2	2	100%	0
premarin	2	2	100%	0
fezolinetant	1	1	100%	0

overall named-entity recall 95.6% · mentions whose name is absent from the text 30

Insights and Findings

